

Cyber Security

IOT $\rightarrow$ internet of things $\rightarrow$ anything connected to internet
ex:- CCTV, mobile, laptop etc..

What is network?

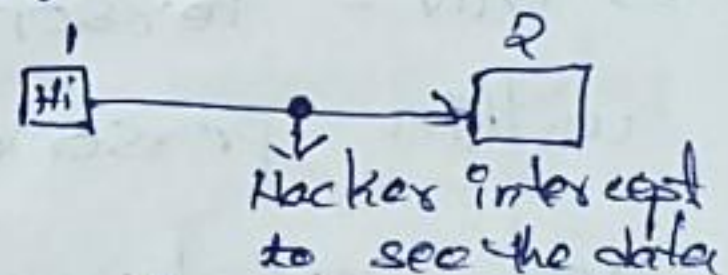
Networking enables computers to communicate and share resources through interconnected systems using standardized protocols and addressing schemes.

• Protocols means rules that internet follows to work.

What is use of networking in cyber security?

it is used to following things.

$\rightarrow$ Packet Sniffing & Analysis:- intercept and decode network traffic to identify vulnerabilities and sensitive data transmission



$\rightarrow$ Protocol Exploitation:- understand weakness in TCP/IP stack to execute attacks like session hijacking or DNS poisoning

$\rightarrow$ Network Reconnaissance; - means - information gathering about target. Map target infrastructure using tools like Nmap to discover host, open ports, and services running.

$\rightarrow$ Firewall Evasion:- Bypass security controls by manipulating packet headers and understanding filtering mechanisms

Types of Networks

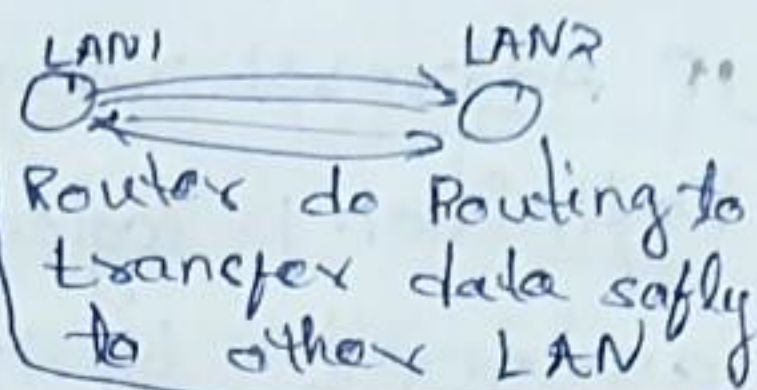
- LAN - Local Area Network; - connects devices within limited geographical area like office building or campus. It is very secure because of small range.
- WAN - Wide Area Network; - spans large distances across cities, countries, or continents connecting multiple LANs
- MAN - Metropolitan Area Network; - covers city or large town area, larger than LAN but smaller than WAN
- PAN - Personal Area Network; - connects devices within personal workspace like smartphone, laptop and peripherals.

* Fiber optic cables

They are the cables that install under water and connected to other countries to communicate or data transfer.

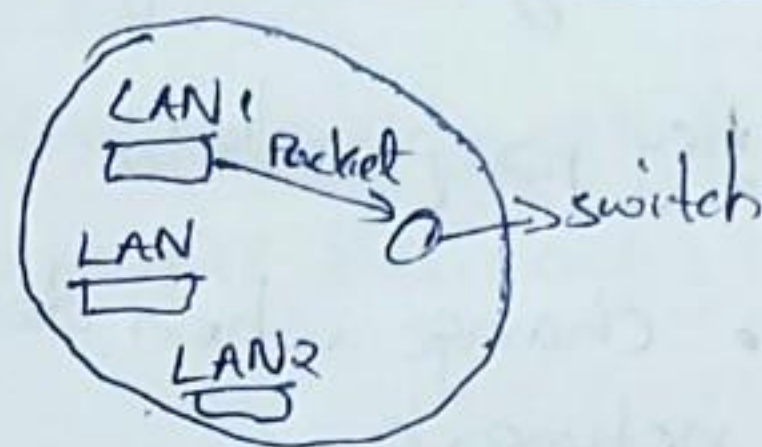
Network Devices

→ Router :- Forwards packets between different networks using IP addresses



→ Switch :- connects devices on same networks using mac addresses.

→ Switch discover data belongs to which Mac address and transfer data



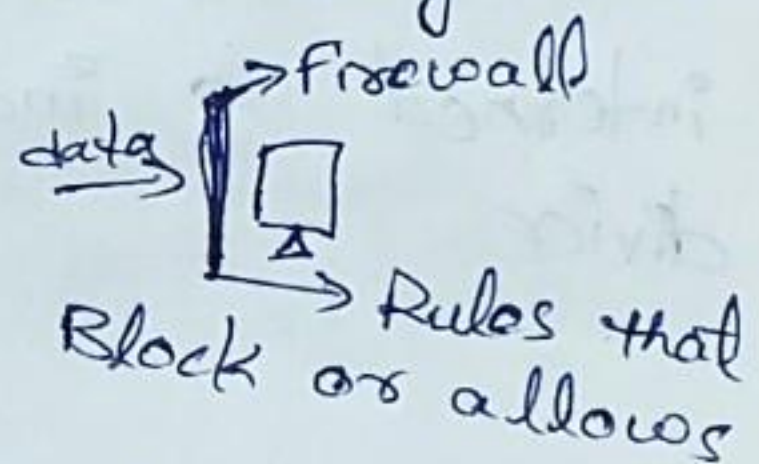
→ Hub :- Broadcasts data to all connected devices (obsolete technology)

- Hub transfer data to all connected devices without using mac addresses
- It is not used in present time because it transfer data to all



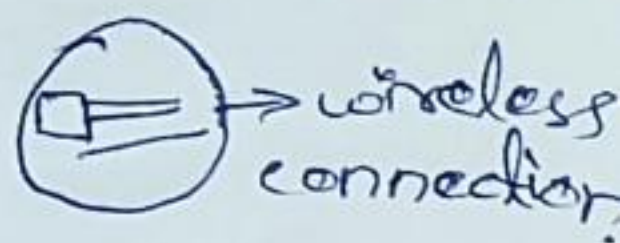
→ Firewall :- Filters traffic based on security rules blocking unauthorized access.

→ To protect the data from the hackers



→ Modem :- converts digital signals to analogue for internet connectivity

→ Access Point :- Provides wireless network connectivity to devices.



IP Address vs MAC Address

IP Address: Internet Protocol
Definition: - logical address assigned by network.

Key properties:

- change when device moves networks.

- Assigned by network administrator or DHCP.

DHCP is a Protocol that ~~assign~~ automatically assign an IP addresses

- Used for routing across networks

- It is the address in internet to identify our device

MAC Address: Media access control address.

Definition: - Physical address burned into hardware

Key properties:

- Permanent and unique to device

- Never change

- Used for local network communication.

* IP addresses has two versions they are.

1. IPV4

2. IPV6

IPv4 vs IPv6 Comparison

IPv4

Format: 32-bit, dotted decimal notation.

Example: 192.168.0.1
 ↓ ↓ ↓ ↓
 8-bit 8-bit 8-bit 8-bit

Structure breakdown into 4 octets (8 bits each)

192.168.0.1

Range is from 0-255
the IP address will be between
this number only.

The new version IPv6

Format: 128-bit, hexadecimal notation

Example: 2001:0db8::1

Structure breakdown into 8 groups (16 bits each)

2001:0db8:0000:0000:
0000:0000:0000:0001

→ IPv4 has $2^{32} \rightarrow$ 4.8 billion addresses but would
population is 8 billion so IP are less, so
New version of IP is created 2^{128}

→ But ⁱⁿ present time we use IPv4 only
because ^{IPv6} ~~is~~ is not compatible as IPv4.

Public vs Private IP Addresses

Public IP

Defination:- Globally unique address routable on internet

Example:- 8.8.8.8 (Google DNS)

Use Case:- servers, devices accessible from internet.

→ it will communicate only with public IP to Public IP

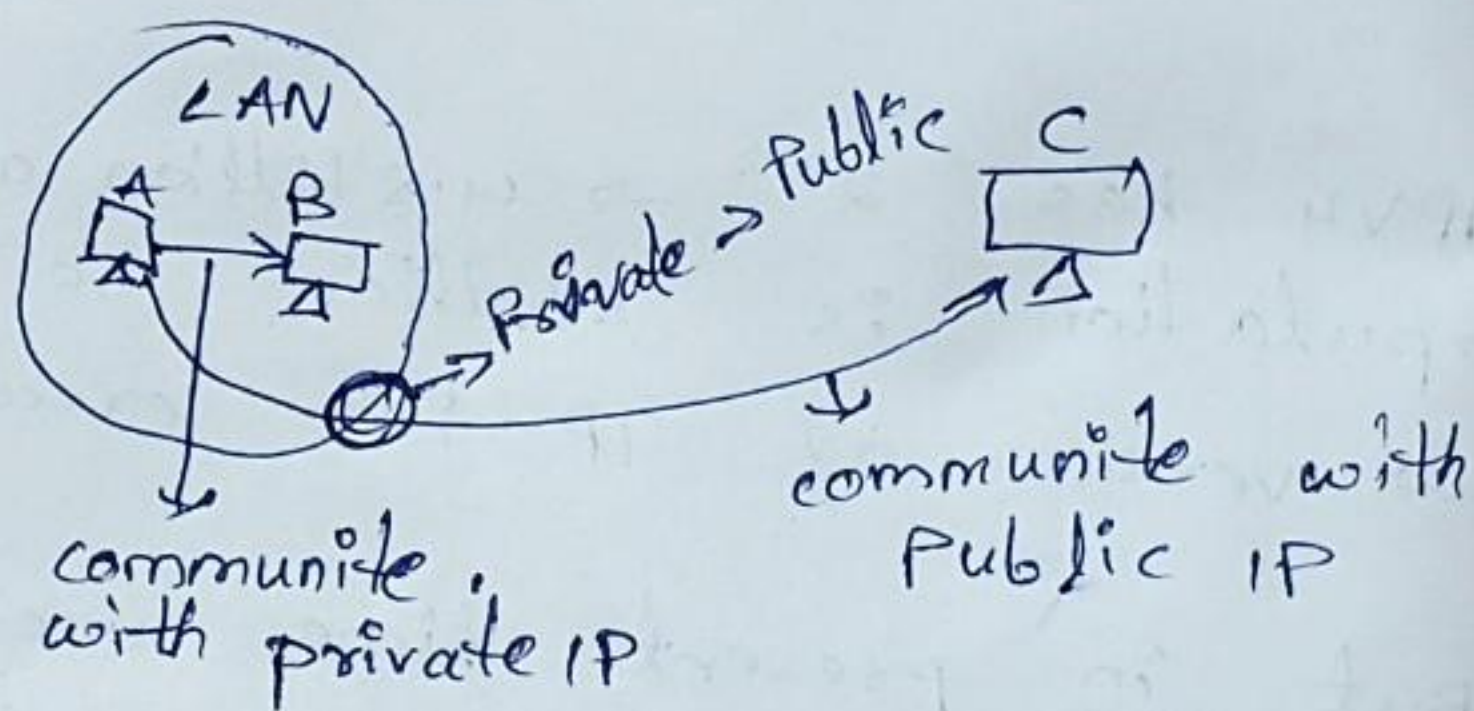
Private IP

Defination:- Non-routable address used within private networks

Example:- 192.168.0.12

Use Case:- Home network, corporate LANs, internal devices.

→ it will communicate only with private IP to Private IP



→ Private IP is converted into Public using Router which is called NAT.

Private IP ranges:-

- 10.0.0.0 to 10.255.255.255 (class A)

- 172.16.0.0 to 172.31.255.255 (class B)

- 192.168.0.0 to 192.168.255.255 (class C)

Static vs Dynamic IP Assignment

Static IP

Definition: Manually assigned fixed address

characteristics:

- Does not change
- Assigned by administrator
- Used for servers, printers, network devices.

Example:

8.8.8.8 (Google DNS)

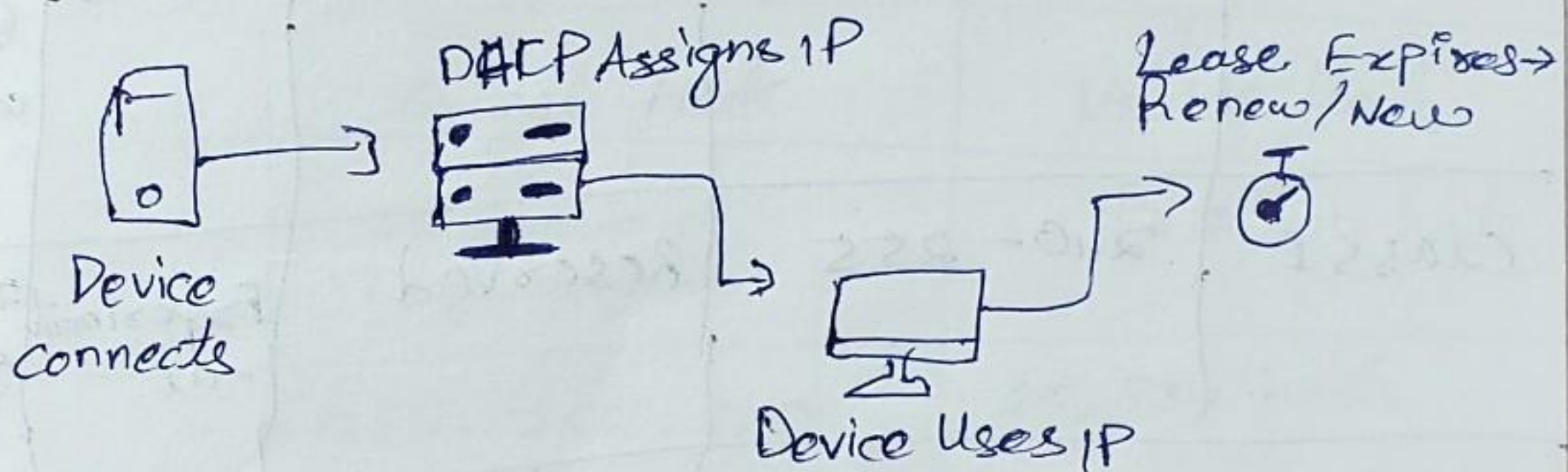
• it is expensive to ~~to~~ buy

Dynamic IP

Definition: Automatically assigned by DHCP server

characteristics:

- change periodically or reconnecting.
- Assigned by DHCP protocol
- Used in home networks mobile devices.
- it is free auto ~~generated~~ generated by DHCP server.



- It generate free IP address. If we disconnect and reconnect IP address will be changed and a new IP is ~~add~~ assigned.

IP Address classes (classful Addressing)

Definition: Original method of dividing IPv4 address into classes based on first octet.

class	First Octet Range	Network Bits	Host Bits	Example.
class A	1 - 255	8-bits	24 bits	10.0.0.0 (16 million host)
class B	128 - 191	16 bits	16 bits	172.16.0.0 (65,000 hosts)
class C	192 - 233	Reserved 24 bits	Multicast 8 bits	192.168.0.0 (254 hosts)
class D	224 - 239	Reserved	multicast	224.0.0.0 (group communication)
class E	240 - 255	Reserved	Experimental	240.0.0.0 (research only)

Example

192.168.11.10

class A
192.168.11.10
↓ ↓
Network host
Bits Bits

class C is mostly used in Home Networks
Ex: 192.168.11.10.

Network bits [192.168.11] [10] → host bits

Key Points

- class A = Large networks (few networks, many host)
- class B = Medium networks (Balanced)
- class C = small networks (many networks, few host)
- class D & E = special use, not for standard addressing

→ classless inter domain routing

CIDR Notation

A way to express IP address and their network size in one line.

Network Portion.

First 24 bits - identifies the network.

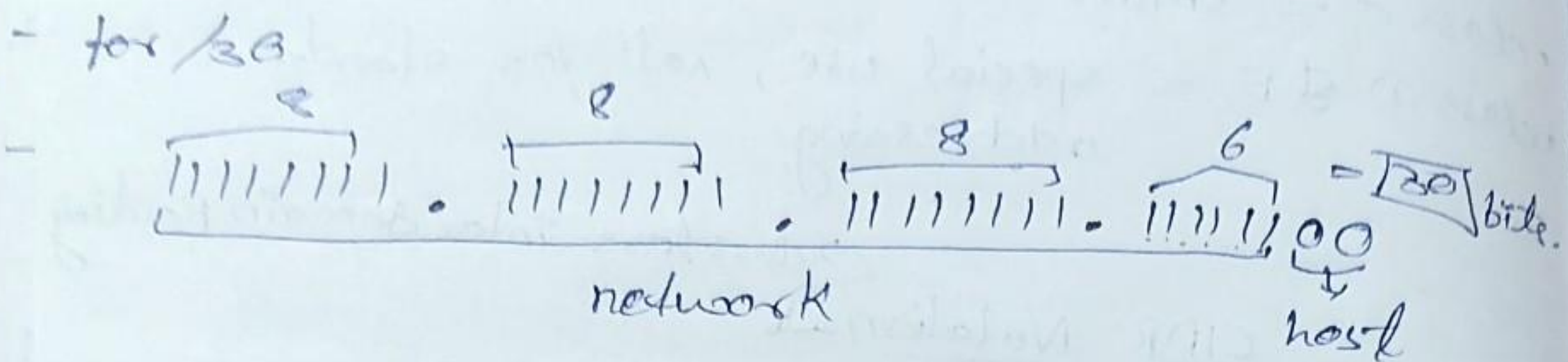
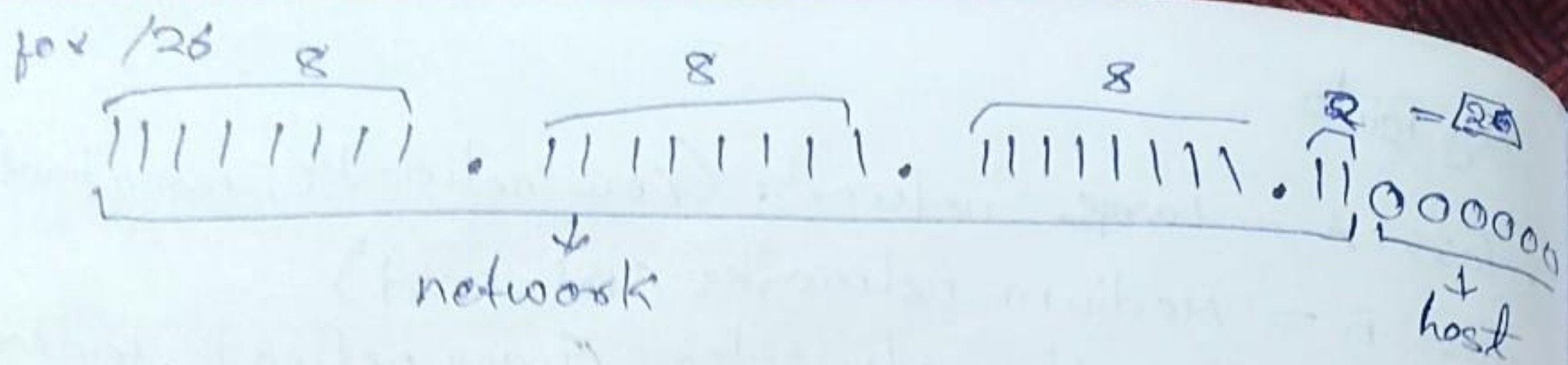
Example: 192.168.1

Host Portion

Last 8 bits - identifies the device

Example: 0 to 255.

CIDR	Subnet Mask	Usable Hosts.
/8	255.0.0.0 Large network	16,777,214 hosts
/16	255.255.0.0 medium network	65,534 hosts
/24	255.255.255.0 small network	254 hosts
/26	255.255.255.192	62 hosts
/30	255.255.255.252	2 host.



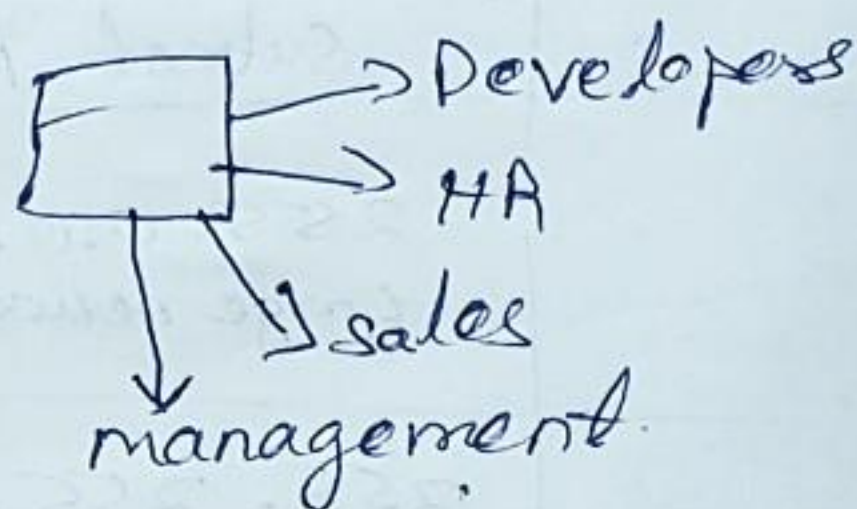
they are used to subnetting. 1261/30

* What is ~~a~~ subnetting?

• Dividing one large network into smaller, manageable pieces.

The Big Idea Example

192.168.10/24 - one Big Network.



giving For a different ~~giving~~ Range of IP to every department to maintain security.

for dividing Big Network.

After splitting:

subnet A	subnet B	subnet C	subnet D
192.168.10/26	192.168.1.64/26	192.168.128/26	192.168.1.192/26
Hosts: 62	Hosts: 62	Hosts: 62	Hosts: 62

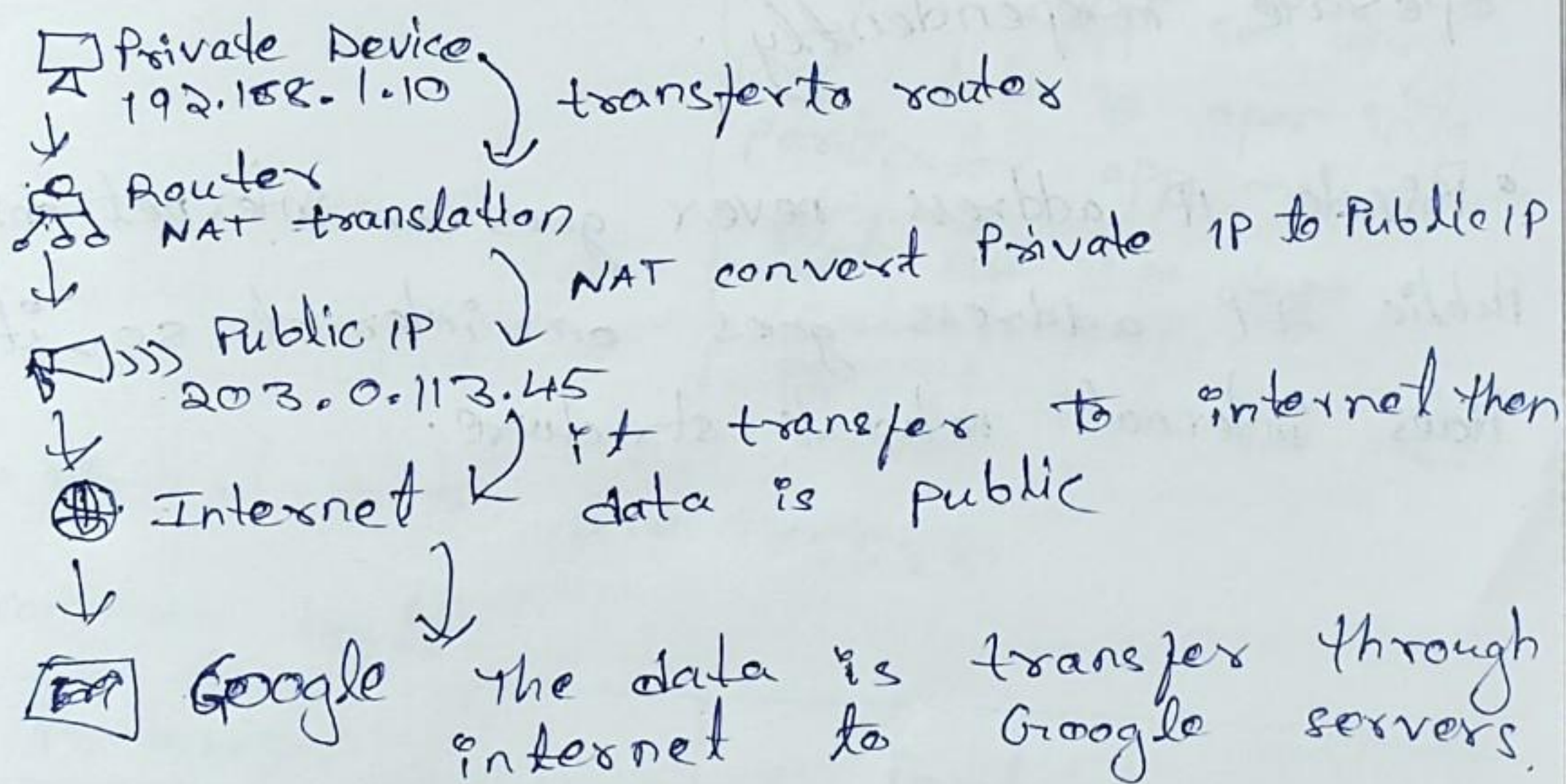
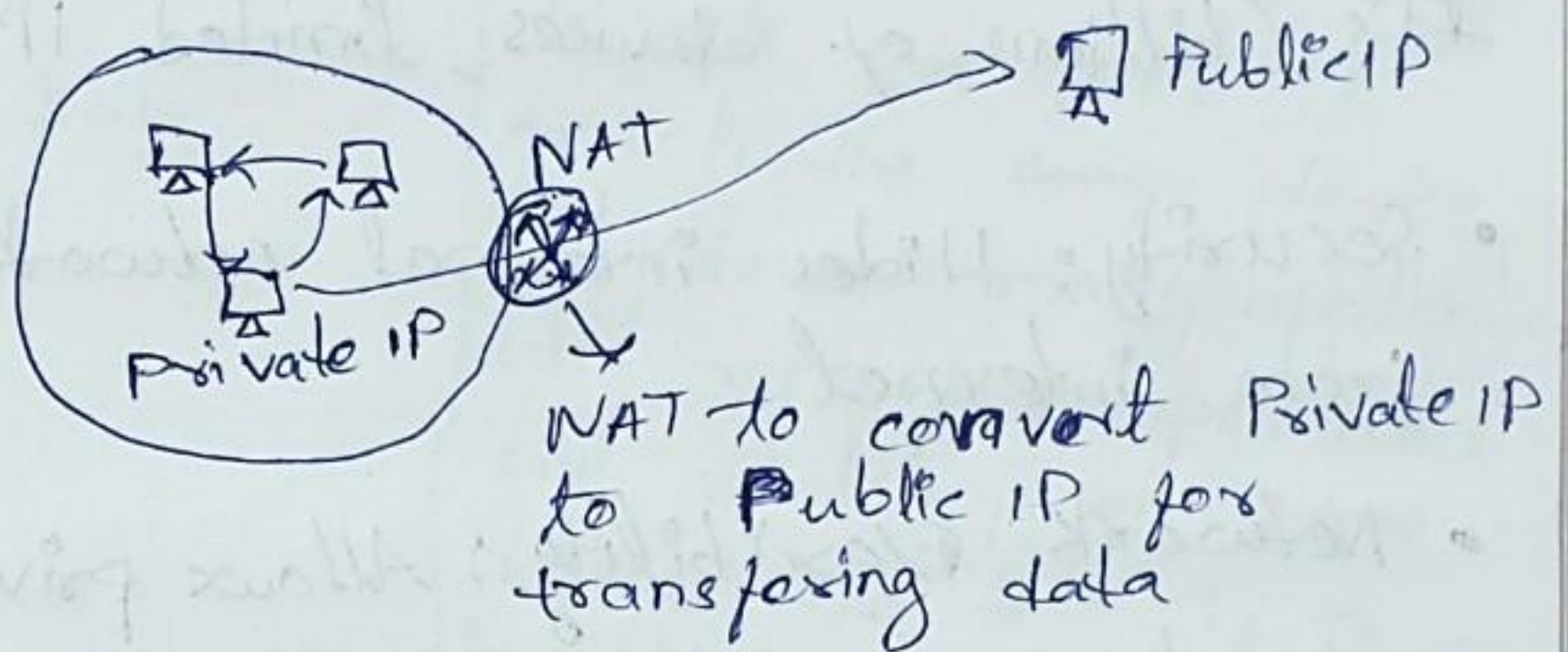
Each department will get 62 IP's

* NAT - Network Address Translation.

NAT (Network Address Translation): Allows private IP addresses to communicate with the internet by translating them to public IP addresses.

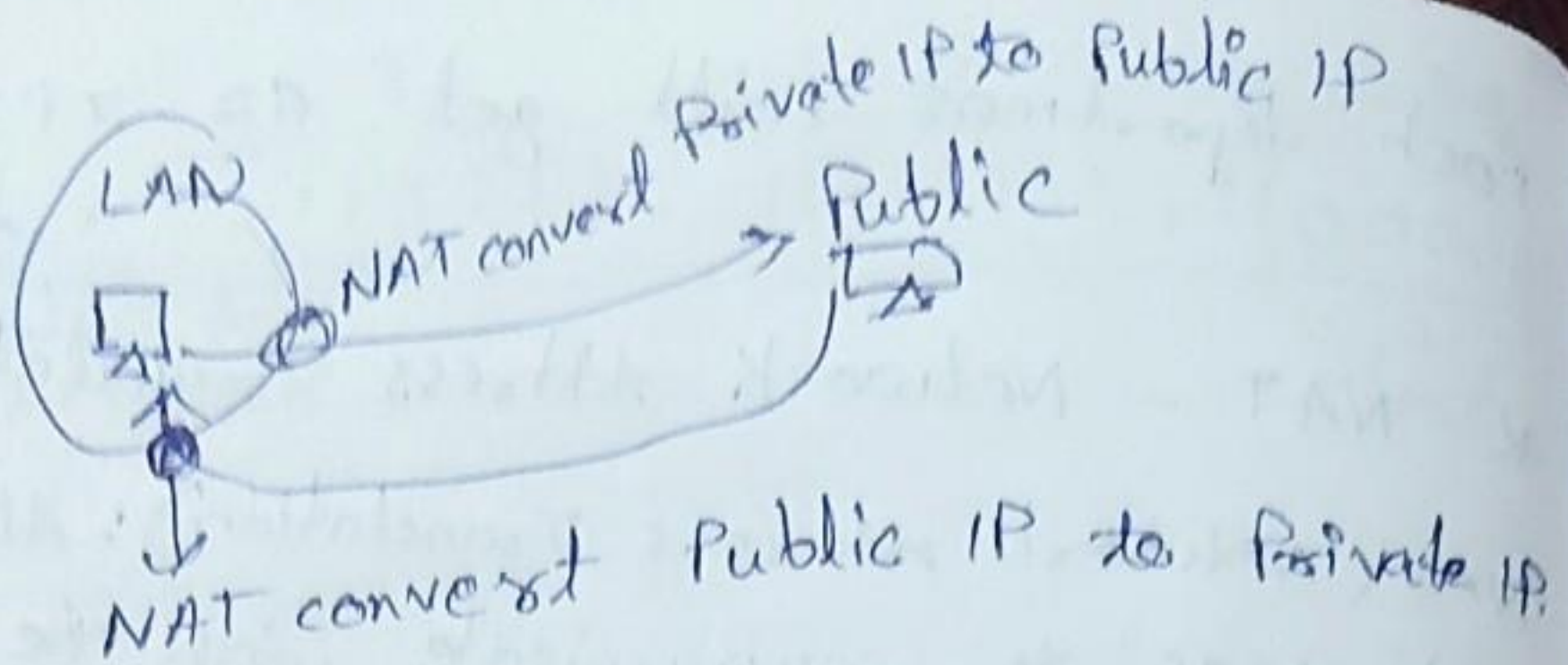
(or)

It converts private IP addresses to Public IP address.



Translation Step (at Router):

- Private IP replaced by public IP for outbound traffic
- Response traffic translated back to private IP for device.



Why NAT Exists:

- IP Address Conservation: Reduces need for public IPs (billions of devices, limited IPv4 address)
- Security: Hides internal network structure from internet.
- Network Flexibility: Allows private networks to operate independently.
- Private IP address never goes on internet only Public IP address goes on internet so, it hides internal network structure.

What is a Protocol and Port?

↓
Rules and regulation

Protocol

Rules and standards for communication between devices

Ex:

HTTP, HTTPS, FTP, DNS, SSH, SMTP, SSL

Port:

Logical endpoint or "door" on a device for network communication

Range: 0 - 65535

(65536 total ports)

Example use IP's to reach the computer device but in devices many applications will run if I open mail and whatsapp on same device the data is to open in which app so, using port data is open into particular app or website.

Port are two type
Virtual ports

Physical ports like USB etc..

• Virtual Ports are 65535

Common Ports.

Protocols	Port.
HTTP	80
HTTPS	443
FTP	21
SSH	22
DNS	53

Port Ranges:-

0-1023 = well known ports.

1024-49151 = Registered Ports.

49152-65535 = Dynamic / Private Ports.

↓
Use by Device to share data

Network foundation:-

OSI Model - All 7 Layers.

OSI → (open system Interconnection)

Layer 7 - Application

User interface and network services
uses protocols like HTTP, HTTPS, DNS, SSL are application layer protocols.

Layer 6 → Presentation;

converting data into compact form like
10MB → 5MB

Data translation, encryption, compression.

↓
means converting data into unreadable form.

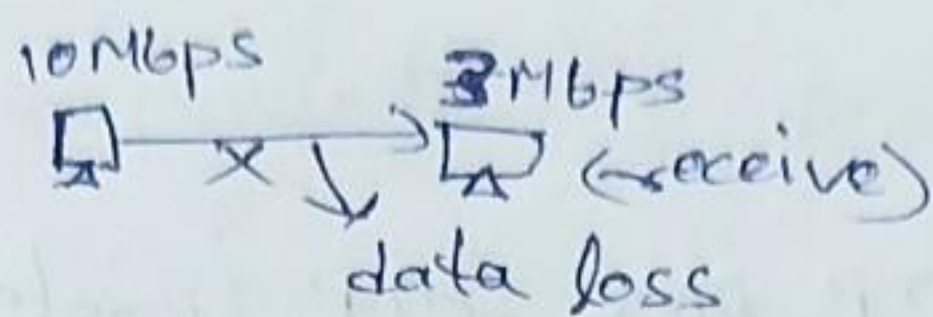
Layer 5 → Session

If two devices are communicating a session is created to manage that session we use session layer.

• Establishes and manages connection.

TCP Layer 4 - Transport

TCP / UDP, error recovery, flow control
→ end to end reliable delivery of data



→ It manage manages flow ~~and~~ to avoid data loss

Layer 3 - Network

IP addressing and routing packets.

→ If our device want to share data with other device it choose has many roots so, it decide which root is better to share data.

Layer - 2 - Data link.

MAC addressing and framing

→ It provides mac address (Physical Address)

Layer 1 - Physical

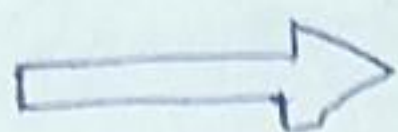
Physical transmission of bits

It manages Physical connections

→ OSI model is made for conceptual model
it is made for only understanding

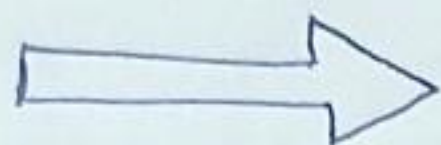
→ We use in daily internet is TCP/IP Model.

TCP/IP Model Explained.



Application Layer

It uses HTTP, FTP, DNS, SMTP protocols



Transport Layer

TCP, UDP for reliable data delivery



Internet Layer → Network

IP, ICMP, ARP for addressing and routing



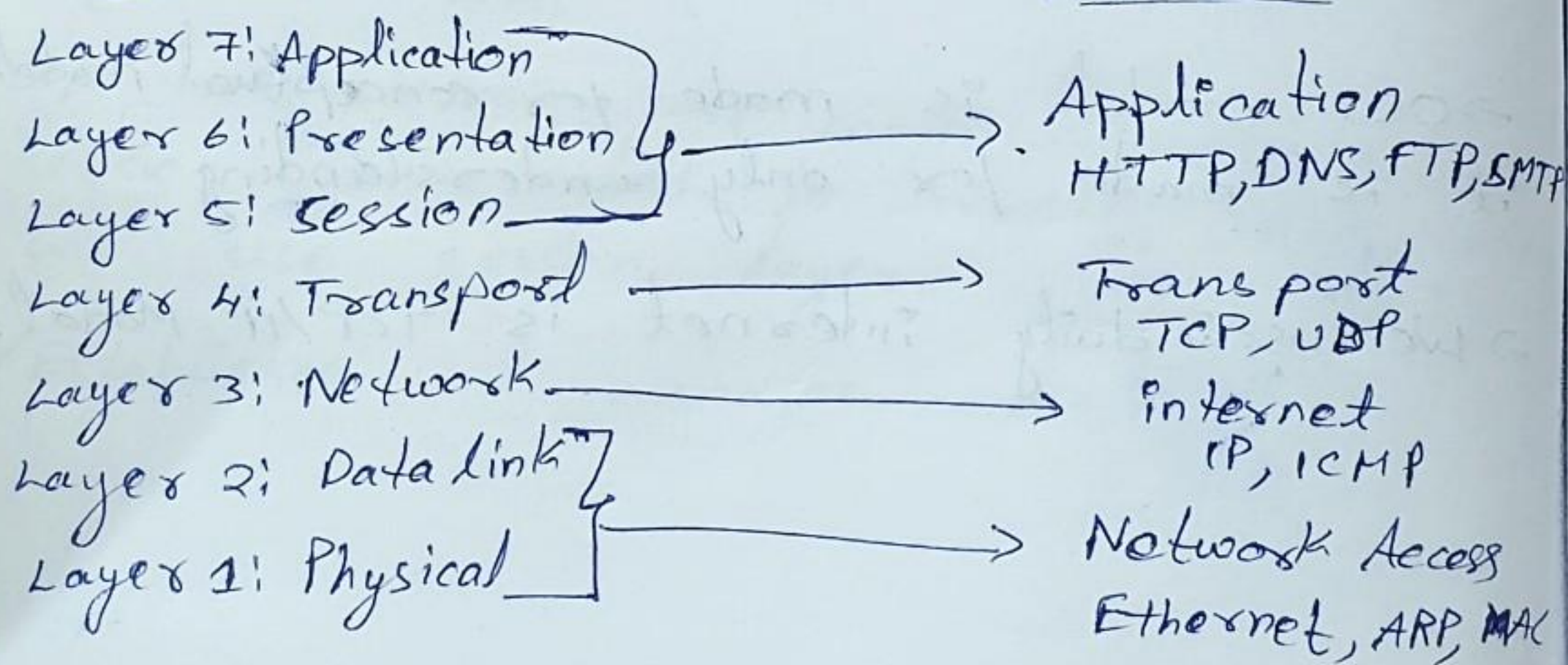
Phy. Network Access layer

Physical hardware and data link protocols

OSI VS TCP/IP comparison

OSI Model

TCP/IP Model



* The TCP/IP model consolidates OSI's top three layers into a single application layer and combines the bottom two into Network access layer.

Data Encapsulation & Decapsulation

Data is shared on Application Layer 

 Segment
It divide data into segment
Transport Layer adds
TCP / UDP header

Data is kept in. Packet 
Network Layer adds
IP header

 Frame
Data link Layer adds
mac addresses

Bits 
Physical Layer
transmits signals

Encapsulation wraps data with headers at each layer. Decapsulation reverses the process when receiving.

TCP vs UDP Comparison

Feature	TCP	UDP
connection	connection-oriented	connection less
speed	slower (reliable)	faster (no overhead)
Reliability	Guaranteed delivery	No guarantee
Error checking	Yes, extensive	Minimal
ordering	Maintains packet order	No ordering
Use case	web, Email, file Transfer	Gaming, streaming, DNS

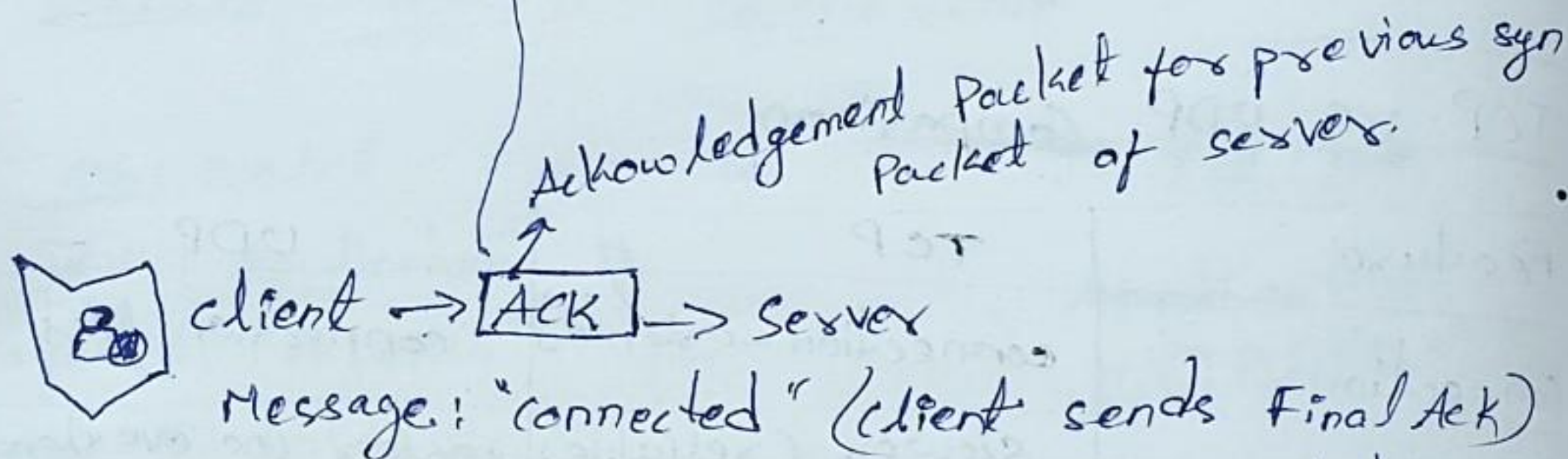
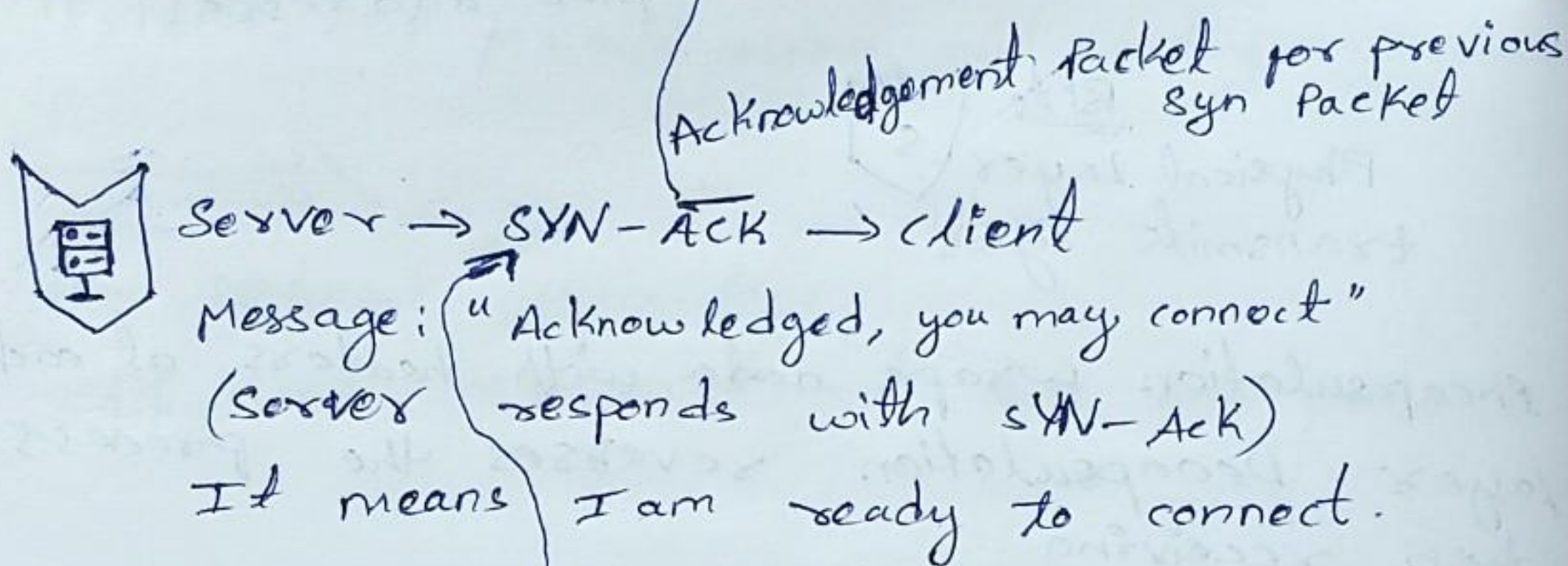
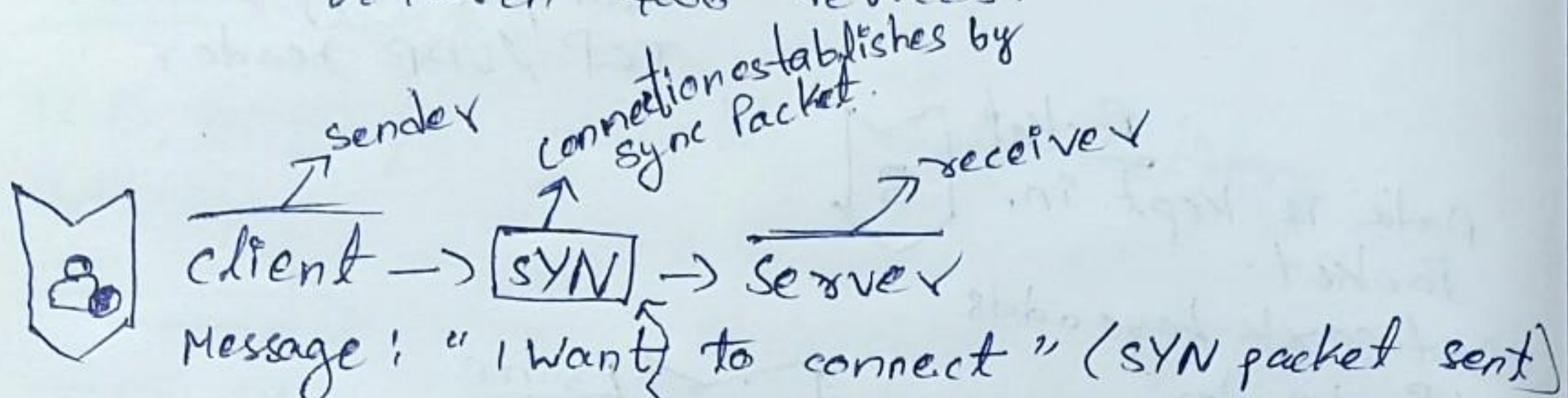
Key Differences

TCP = Reliable, ordered, connection-based

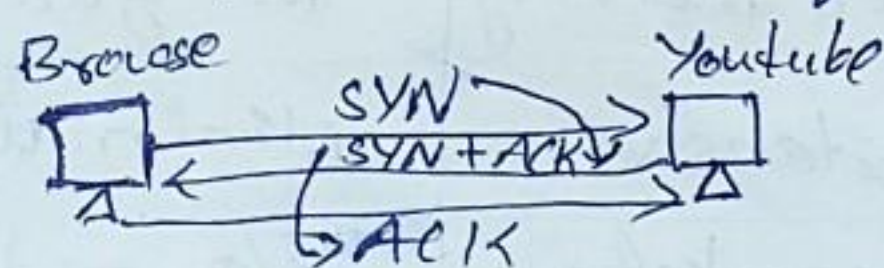
UDP = Fast, connectionless, best-effort

TCP 3-Way Handshake

Definition:- Establishes a reliable TCP connection between two devices.



If you open Browser for running youtube



Connection establishes

TCP 4-Way Handshake (Connection Termination)

Definition: Gracefully terminates a TCP connection between two devices.

client → FIN → server
Message: "I'm done sending data" (FIN packet is sent)

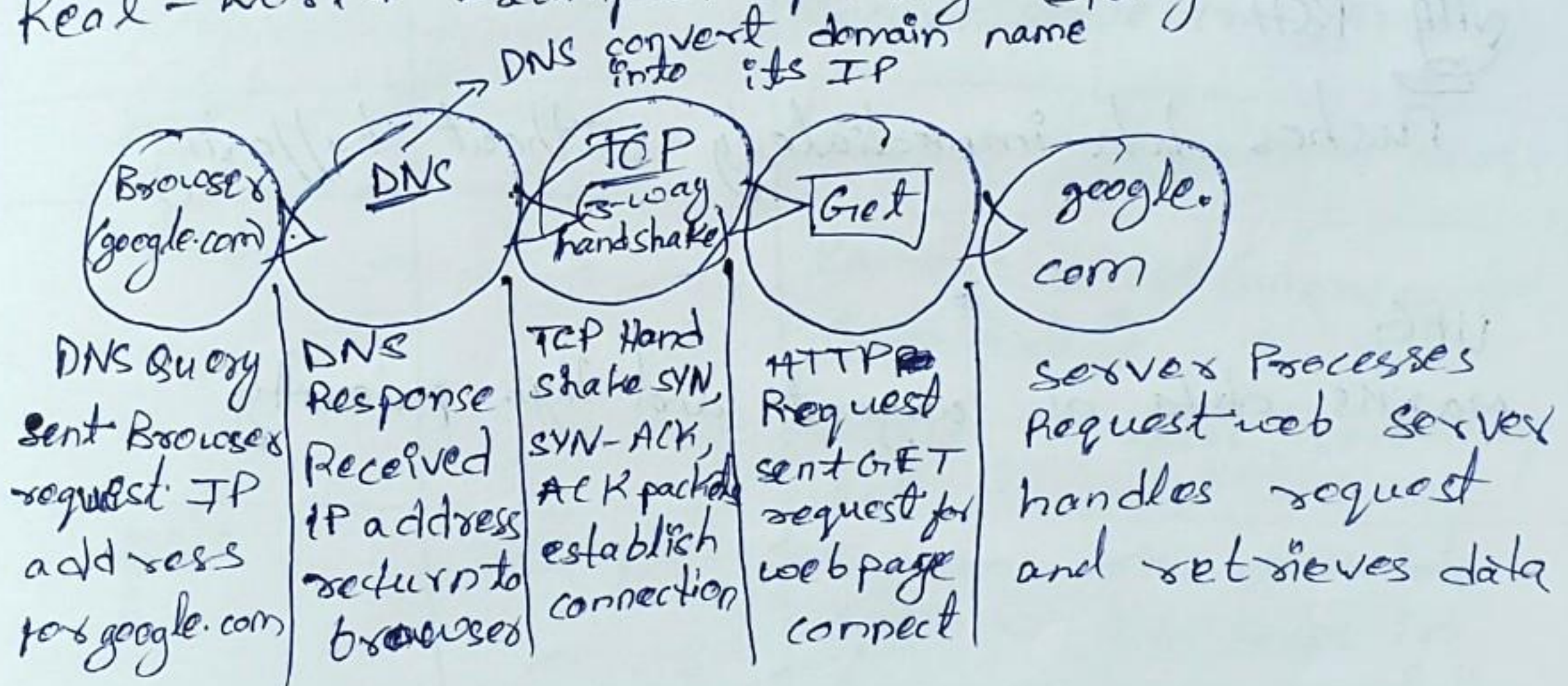
server → ACK → client
Message: "Got it, I acknowledge." (server sends ACK)

server → FIN → client
Message: "I'm done too" (server send its own FIN)

client → ACK → server
Message: "Confirmed, connection closed"
(client sends final ACK)

"Connection closed."

Real-World Example: Opening Google.com



Common TCP Flags

TCP flags control the state and behavior of communication

 SYN

Initiates a new connection (synchronization)

 ACK

Acknowledges receipt of data or packet

 FIN

Gracefully closes a connection

 RST

Forces an immediate connection reset

 PSH

Pushes data immediately without buffering

URG

Marks data as urgent and high priority.

Common Protocols Quick Reference

Protocol	Port	Purpose
HTTP	80	Unencrypted web traffic • Hyper Text Transfer Protocol Unencrypted mean data transfer in same form <u>Example</u> → Hi send in same form → Hi
HTTPS	443	Encrypted web traffic (SSL/TLS) • Hyper text Transfer Protocol Secure. Encrypted mean data transfer into cipher language
FTP	21	File transfer Protocol (unencrypted)
SFTP	22	Secure file transfer (SSH)
SSH	22	secure shell, remote access
Telnet	23	Remote access (unencrypted, deprecated) Telnet is older version of SSH.
DNS	53	Domain name resolution system. It helps in Domain name resolution.
DHCP	67-68	Dynamic IP assignment → automatically assign IP to devices.
SMTP	25	send mail.

POP3	110	Receive email (old)
IMAP	143	Receive email (modern)
RDP	3389	Remote desktop (Windows)

Encryption & Decryption Process

Definition:-

converting plain text into ciphertext (encryption) and back (decryption) using decryption keys.

Encryption

Step 1:- Plain text (Readable data)
Encryption Key.

Step 2:- Encryption Algorithm
(AES, RSA, DES)

Step 3:- Ciphertext (unreadable
scrambled data)

Result:- Data is now secure and
unreadable without key

Decryption

Step 1:- Ciphertext (encrypted data)
* Decryption.

Step 2:- Decryption Algorithm
(reverse of encryption)

Step 3:- Plain text (original
readable data)

Result:- Data is recovered
and readable again

Symmetric Encryption:-

Same key for encryption and decryption (AES, DES)

- Fast, efficient
- Key must be shared securely
- Used for bulk data

Asymmetric Encryption:

different key.

- slower but secure
- Public key encryption and Private key decrypt
- mostly used in key exchange and digital signature

DNS - How Domain Resolution Works (Domain Name System)

1 user types google.com in browser

↓
query sent to Recursive Resolver (ISP DNS)

↓
Resolver queries Root Name server (root)

↓
Root directs to TLD server (com) (Top level domain)

↓
TLD server directs to Authoritative Name server.

↓
Authoritative server returns IP address
(142.250.195.46)

↓
Browser connects to IP and loads website.
